Continuous Compounding & Euler's Constant e

Exponential & Logarithmic Functions • Standard: CCSS.MATH.HSF.LE.A.4

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1. MATHEMATICAL CONCEPT & STACK APPROACH: Evaluate discrete and continuous exponential growth models from the inside out (exponent first, then base power, then initial value).

Exponential growth P·e^(rt) compounds continuously by scaling a base quantity according to an inner rate-time product (r × t). Evaluating from the inside out—computing the exponent r × t first, raising the base, and multiplying by P—follows the exact mathematical hierarchy without parentheses.

2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

Published Block Kit: Prime Factor Studio (PRIME-01 Logarithmic Blocks) & Growth Curve

Hands-On Setup: Stack Prime Factor [2] blocks (15 mm each) to show linear height addition representing exponential doubling ($2^1 = 2$, $2^2 = 4$, $2^3 = 8$).

Student Prediction (Before Calculator): Invest $1,000 at 7% continuous interest for 10 years: A = 1000 × e^(0.07 × 10) = 1000 × e^0.7. Will it double? (e^0.7 ~ 2.01).

3. CALCULATOR EXECUTION (SC-32 KEYCAPS & 4-LEVEL STACK TRACE)

Keystroke Flow: [0] [•] [0] [7] [ENTER] [1] [0] [×] [e^x] [1] [0] [0] [0] [×]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[0] [•] [0] [7]	0.07	Rate r = 0.07 entered in X
[ENTER]	0.07	Rate lifted to Y
[1] [0]	10	Time t = 10 entered in X
[×]	0.7	Exponent r × t = 0.70
[e^x]	2.0138	e^0.70 = 2.01375 (Growth factor: more than doubled!)
[1] [0] [0] [0]	1000	Principal P = 1000 (growth factor lifts to Y)
[×]	2013.75	Total accumulated amount A = $2,013.75!

4. MATHEMATICAL CONTEXT & REAL-WORLD MOTIVATION (John Napier's Logarithms: Turning Multiplication into Height Addition)

In 1614, Scottish mathematician John Napier realized that logarithms turn tedious multi-digit multiplication into simple addition—the exact engineering principle behind our Prime Factor Studio blocks, whose physical heights scale as h = 15 × log2(p) mm so stacking blocks adds their logarithms! Paired with Jakob Bernoulli's continuous compounding constant e, logarithms unlock doubling time and half-life.

Core Investigation (Student Warm-Up A):

A bacteria culture starts with P = 500 cells and triples every hour (A = 500 × 3^t). Calculate the population after t = 4 hours.

Solving for Time: Doubling Time, Half-Life & Natural Logarithms

Logarithms & Inverting Exponentials • Standard: CCSS.MATH.HSF.BF.B.5

1. EXTENSION CONCEPT (WARM-UP B), BLOCKS & PREDICTION

Concept & Stack Approach: A logarithm is the inverse of exponentiation: it converts multiplicative scaling into additive height (ln(a·b) = ln(a) + ln(b)). To solve for time t in P·e^(rt) = A, divide A by P first, take the natural logarithm ln(A/P) to unwrap the exponent, and divide by r.

Blocks & Prediction: Observe how Prime Factor block heights scale with ln(p) on the comparator lid while comparing additive block heights to multiplicative products. Prediction: At 5% continuous growth (r = 0.05), doubling time is t = ln(2) / 0.05 ~ 0.69315 / 0.05 ~ 13.86 years.

2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

[2] [LN] [0] [•] [0] [5] [÷]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[2]	**2**	Doubling target factor 2
[LN]	**0.6931**	Natural log ln(2) = 0.693147... in X
[0] [•] [0] [5]	**0.05**	Rate r = 0.05 (ln(2) lifts to Y)
[÷]	**13.86**	t = ln(2) / r = 13.86 years!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Multiplying the base by the exponent (computing 2^5 as 2 × 5 = 10) or reversing Base and Exponent on [y^x].

Hands-On Block Remediation: Stack five [2] Prime Factor logarithmic blocks in the comparator lid to show that adding 5 block heights multiplies 2×2×2×2×2 = 32, and verify that Base y goes in Register Y and Exponent x goes in X before [y^x].

Misconception 2: Applying the power [y^x] after multiplying by principal P in P(1 + r)ᵗ (computing (P·(1+r))ᵗ instead of P·((1+r)ᵗ)).

Hands-On Block Remediation: Use the Inside-Out RPN rule: compute (1 + r)ᵗ first ([1.05] [ENTER] [2] [y^x]), then multiply by P ([1000] [×]).

4. EXTENSION MATHEMATICAL INSIGHT

A logarithm is the inverse of exponentiation: it converts multiplicative scaling into additive height (ln(a·b) = ln(a) + ln(b)). To solve for time t in P·e^(rt) = A, divide A by P first, take the natural logarithm ln(A/P) to unwrap the exponent, and divide by r.

Exponential Growth, Compound Interest & Logarithms

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: EXPONENTIALS, CONTINUOUS COMPOUNDING & DOUBLING TIME

- Discrete exponential growth multiplies by a constant factor each period: A = P × b^t (or A = P × (1 + r/n)^(n·t)).
- Continuous exponential growth follows A = P × e^(r·t), where Euler's constant e ~ 2.71828 is the base of natural growth.
- Logarithms invert exponentials (ln(e^x) = x) and turn multiplication into addition: ln(a × b) = ln(a) + ln(b).

Key Sequence: `0` `.` `0` `7` `ENTER` `1` `0` `×` `e^x` `1` `0` `0` `0` `×`

Investigation 1: Evaluate e^2 to 4 decimal places.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 2: Find the continuous doubling time t = ln(2) / r for an investment growing at r = 8% (0.08).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 3: Evaluate 10^3.5 (note that 10^3 < 10^3.5 < 10^4).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 4: Solve 10^x = 500 for x using the base-10 logarithm log10(500).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Paper Steps:

Formula / Algebraic Setup:

Intermediate Step(s):

Simplified Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Unit 10: Exponential Growth, Compound Interest & Logarithms — Answer Key

Complete solutions and SC-32 verification keystrokes for Grades 10–12

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1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

Core Investigation (Warm-Up A): A bacteria culture starts with P = 500 cells and triples every hour (A = 500 × 3^t). Calculate the population after t = 4 hours.

Core Solution: 3^4 = 81; 500 × 81 = 40,500 bacteria.

Core Check Keys: [3] [ENTER] [4] [y^x] [5] [0] [0] [×]

Extension Challenge (Warm-Up B): A radioactive isotope decays according to N(t) = N_0 · e^(-0.035t). Find its half-life t = ln(2) / 0.035 to two decimal places.

Extension Solution: t = ln(2) / 0.035 ˜ 0.6931 / 0.035 ˜ 19.80 years.

Ext. Check Keys: [2] [LN] [0] [•] [0] [3] [5] [÷]

2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

Investigation 1: Evaluate e^2 to 4 decimal places.

Expected Answer: 7.3891

SC-32 Verification Keys: [2] [e^x]

Investigation 2: Find the continuous doubling time t = ln(2) / r for an investment growing at r = 8% (0.08).

Expected Answer: 8.66 years

SC-32 Verification Keys: [2] [LN] [0] [•] [0] [8] [÷]

Investigation 3: Evaluate 10^3.5 (note that 10^3 < 10^3.5 < 10^4).

Expected Answer: 3162.28

SC-32 Verification Keys: [3] [•] [5] [↶] [10^x]

Investigation 4: Solve 10^x = 500 for x using the base-10 logarithm log10(500).

Expected Answer: 2.6990

SC-32 Verification Keys: [5] [0] [0] [↶] [LOG]

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.